Assignment 2

Problem 1

a) $\dot{x}_1 = x_2 = 0$

$$\dot{x}_2 = -\frac{f_3}{m}x_1^3 - \frac{f_1}{m}x_1 - \frac{d}{m}x_2 - g + \frac{u}{m} = -g$$

$(0,0)$ is not an equilibrium point

b) $-\frac{f_3}{m}x_1^3 - \frac{f_1}{m}x_1 - \frac{d}{m}x_2 - g + \frac{u}{m} = 0$

Substitute $(x_1, x_2) = (x_{1d}, 0)$

$$-\frac{f_3}{m}x_{1d}^3 - \frac{f_1}{m}x_{1d} - g + \frac{u}{m} = 0$$

$$u_0 = f_3 x_{1d}^3 + f_1 x_{1d} + mg$$

c) $\tilde{x} = x - x_0$ and $\tilde{u} = u - u_0$, $x_0 = (x_{1d}, 0)$

$$\dot{\tilde{x}} = \dot{x}$$

$$\dot{\tilde{x}}_1 = \tilde{x}_2 + x_{2,0} = \tilde{x}_2$$

$$\dot{\tilde{x}}_2 = -\frac{f_3}{m}\left(\tilde{x}_1 - x_{1,0}\right)^3 - \frac{f_1}{m}\left(\tilde{x}_1 - x_{1,0}\right) - d(\tilde{x}_2 + \overset{0}{x_{2,0}}) - g + \frac{\tilde{u} + u_0}{m}$$

$$= -\frac{f_3}{m}\left(\tilde{x}_1 - x_{1d}\right)^3 - \frac{f_1}{m}\left(\tilde{x}_1 - x_{1d}\right) - d\tilde{x}_2 - g + \frac{\tilde{u} + f_3 x_{1d}^3 + f_1 x_{1d} + mg}{m}$$

$$= -\frac{f_3}{m}\left[\left(\tilde{x}_1 + x_{1d}\right)^3 - x_{1d}^3\right] - \frac{f_1}{m}\tilde{x}_1 - \frac{d}{m}\tilde{x}_2 + \frac{\tilde{u}}{m}$$

$\tilde{u} = 0$

$f(\tilde{x}_1, \tilde{x}_2) = 0 \Rightarrow \tilde{x}_1 = 0$

$\tilde{x}_2 = 0$

d) $A = \frac{\partial f}{\partial \tilde{x}}\Big|_{\tilde{x}=0} = \begin{bmatrix} 0 & 1 \\ -\frac{3f_3}{m}(\tilde{x}_1 + x_{1d})^2 - \frac{f_1}{m} & -\frac{d}{m} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{-3f_3 x_{1d}^2 + f_1}{m} & -\frac{d}{m} \end{bmatrix}$

A is Hurwitz $\Leftrightarrow$ $Re(\lambda_i) < 0 \quad \forall i = 1, \dots, n$

$\det(A - \lambda I) = 0$

$\lambda^2 + \frac{d}{m}\lambda + \frac{3f_3 x_{1d}^2 + f_1}{m} = 0$

$\lambda_{1,2} = -\frac{d}{2m} \pm \frac{1}{2}\sqrt{\left(\frac{d}{m}\right)^2 - 4\frac{3f_3 x_{1d}^2 + f_1}{m}}$

$Re(\lambda_{1,2}) = -\frac{d}{2m} < 0 \Rightarrow$ A is Hurwitz

Theorem 4.7:
$\forall i: Re(\lambda_i) < 0 \Rightarrow \tilde{x} = 0$ is locally A.S

Corollary 4.3:
$\forall i \; Re(\lambda_i) < 0 \Leftrightarrow \tilde{x} = 0$ is E.S locally

Problem 2
a) $\dot{x} = -x^5, \quad V(x) = \frac{1}{2}x^2$

Strict L.F $\Rightarrow$ A.S origin

$\dot{V}(x) = \langle \nabla V, f(x) \rangle = -x^6$
$\dot{V}(0) = 0$ and $\dot{V}(x) < 0 \quad \forall x \neq 0 \Rightarrow$ locally A.S

Theorem 4.2
If Strict L.F and V is radially unbounded $\Rightarrow x = 0$ globally A.S

Since $V(x) = \frac{1}{2}x^2 \to \infty$ as $|x| \to \infty$
$x = 0$ is globally A.S

b) $\dot{x}_1 = -x_1 + 4x_2^2$

$\dot{x}_2 = -x_2^3$

$V(x) = \frac{1}{2}(x_1^2 + a_2 x_2), \quad a_2 > 0$

$\dot{V} = x_1\dot{x}_1 + a_2 x_2 \dot{x}_2 = -x_1^2 + (-1+2a_2)x_1 x_2^2 - a_2 x_2^4$

Choose $a_2$ to cancel mixed term:

$-1 + 2a_2 = 0 \Rightarrow a_2 = \frac{1}{2}$

$V = \frac{1}{2}x_1^2 + \frac{1}{4}x_2^2, \quad \dot{V} = -x_1^2 - \frac{1}{2}x_2^4 < 0 \quad \forall x \neq 0$

Theorem 4.1: $x=0$ is A.S

$V \to \infty$ as $\|x\| \to \infty \Rightarrow V$ is R.U

Theorem 4.2: $x=0$ is G.A.S

c) $V = \frac{1}{2}(x_1^2 + a_2 x_2^2), \quad a_2 > 0$

$\dot{V} = x_1(-x_1 + 4x_2) + a_2 x_2(-x_2^3) = -x_1^2 + 4x_1x_2^2 - a_2 x_2^4$

Complete square

$= -(x_1 - 2x_2^2)^2 + (4 - a_2)x_2^4$

$a_2 = 5 \Rightarrow \dot{V} < 0 \quad \forall x \neq 0$

T.4.1: A.S

T.4.2: $V \to \infty$ as $\|x\| \to \infty$

Therefore origin is G.A.S

d) $\dot{x}_1 = -x_1 + x_2(1-x_1^2)$
$\dot{x}_2 = -(x_1 + x_2)(1-x_1^2)$

$V = \frac{1}{2}(x_1^2 + x_2^2)$, $\dot{V} = x_1\dot{x}_1 + x_2\dot{x}_2 = -x_1^2 - x_2^2(1-x_1^2)$
For $|x_1| < 1$ $\dot{V} < 0$ $\forall x \neq 0$

T.4.1 : (0,0) is L.A.S

Equilibrium points : $(0,0)$, $(\sqrt{2}, -\sqrt{2})$, $(-\sqrt{2}, \sqrt{2})$
Therefore $(0,0)$ can't be G.A.S

Problem 3
a) $\dot{x}_1 = x_1^2 - x_2^2$
$\dot{x}_2 = 2x_1x_2$

From the phase portrait we can see that there is no $\delta$ for every $\varepsilon > 0$ such that $\|x(0)\| < \delta \Rightarrow \|x(t)\| < \varepsilon$, $\forall t \geq 0$.
Therefore it is unstable

b) $\dot{x} = \alpha x$, $\alpha < 0$

$x(t) = x(0)e^{\alpha t}$

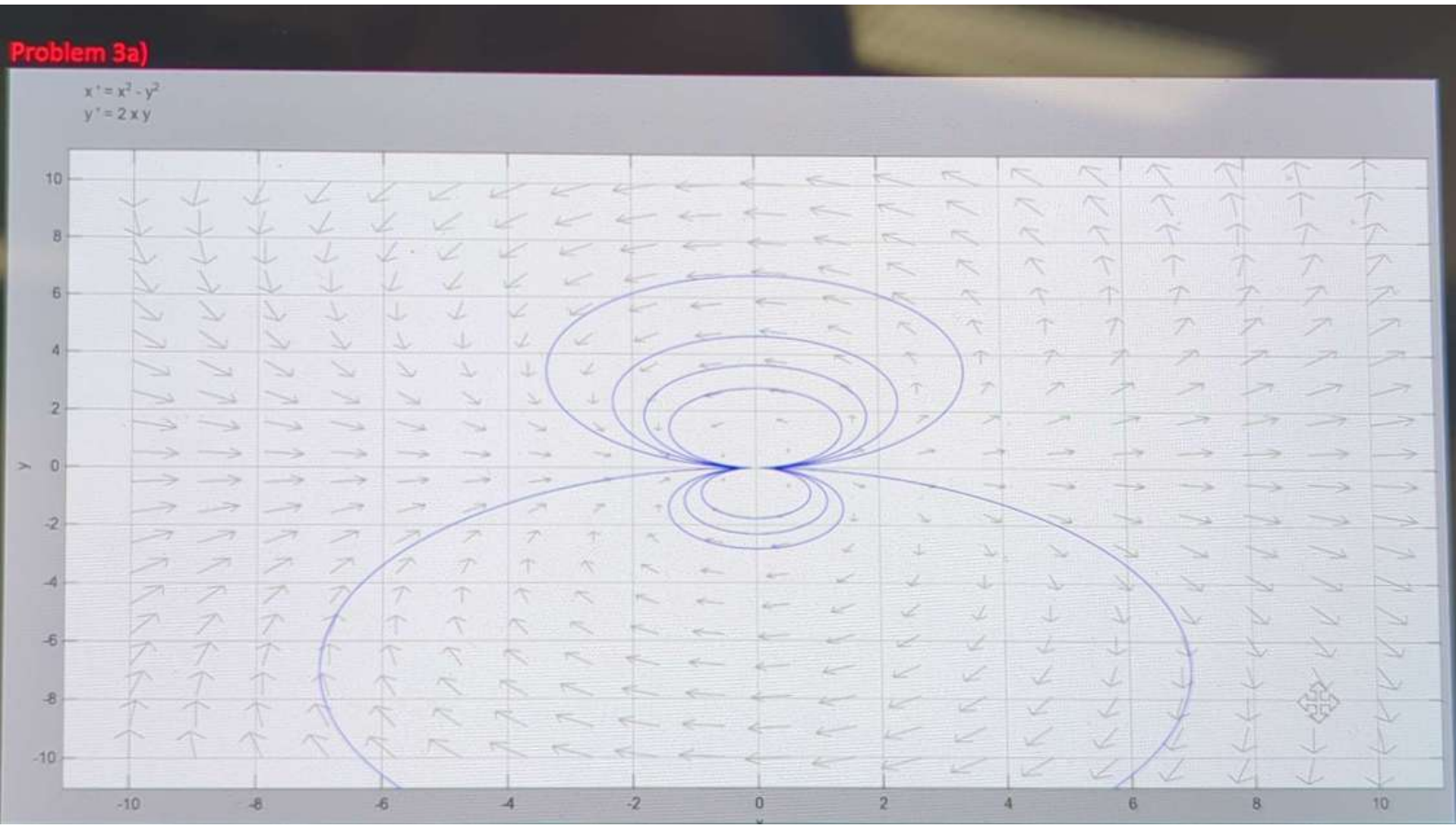
For any $\varepsilon > 0$, choose $\delta = \varepsilon$

If $|x(0)| < \delta$, then since $e^{\alpha t} \leq 1$ for $t \geq 0$,
$|x(t)| = |x(0)|e^{\alpha t} < \delta = \varepsilon$
$\Rightarrow$ Origin is stable

Because $\alpha < 0$, $\lim_{t \to \infty} x(t) = 0$

By Def. 4.1, $x = 0$ is A.S

**Problem 3a)**



Problem 4 a)

$$\dot{x}_1 = x_1^2 - 2x_1x_2 + 2x_1 - x_1^2x_2$$
$$= (x_1^2 + 2x_1) - x_2(x_1 + 2x_1) = (x_1^2 + 2x_1)(1 - x_2)$$

$$\dot{x}_2 = x_1^3 + 2x_1^2 + x_1^2x_2 + 2x_1x_2 = x_1(x_1^2 + 2x_1) + x_2(x_1^2 + 2x_1)$$
$$= (x_1^2 + 2x_1)(x_1 + x_2)$$

b) $x_1^2 + 2x_1 = x_1(x_1 + 2) = 0$

$x_1 = 0$ or $x_1 = -2$ for any $x_2$

$1 - x_2 = 0$, $x_1 + x_2 = 0 \Rightarrow (-1, 1)$

c) $z_1 = x_1 + 1$, $z_2 = x_2 - 1$

$x_1 = z_1 - 1$, $x_2 = z_2 + 1$

$\dot{z}_1 = \dot{x}_1$, $\dot{z}_2 = \dot{x}_2$

$$x_1^2 + 2x_1 = (z_1 - 1)^2 + 2(z_1 - 1) = z_1^2 - 1 = -(1 - z_1^2)$$
$$\dot{z}_1 = (z_2^2 - 1)(-z_2) = z_2(1 - z_1^2)$$
$$\dot{z}_2 = (z_1^2 - 1)(z_1 + z_2) = -(z_1 + z_2)(1 - z_1^2)$$

$$\dot{z} = \begin{bmatrix} z_2 \\ -(z_1 + z_2) \end{bmatrix} (1 - z_1^2)$$

d) $\dot{z} = (1 - z_1^2) A z$, $A = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}$

$Q = I > 0$

Lyapunov equation $A^T P + PA = -Q$

$$P = \begin{bmatrix} \frac{3}{2} & \frac{1}{2} \\ \frac{1}{2} & 1 \end{bmatrix} > 0$$

$V(z) = z^T P z$

$\dot{V} = (1 - z_1^2) z^T (A^T P + PA) z = -(1 - z_1^2) z^T Q z$

for $|z_1| < 1$, $\dot{V} < 0$ $(z \neq 0)$

By Lyapunov's direct method $z = 0$ is L.A.S

Therefore $(x_1, x_2) = (-1, 1)$ is L.A.S.

Problem 5

$V(x) = \frac{1}{2} x^T P x + \gamma (1 - \cos x_1)$, $P = \begin{bmatrix} P_{11} & P_{22} \\ P_{21} & P_{22} \end{bmatrix} > 0$

$\dot{V} = (P_{11} - 2P_{21}) x_1 x_2 + (P_{21} - 2P_{22}) x_2^2 - 10 P_{21} x_1 \sin x_1$
$+ (\gamma - 10 P_{22}) x_2 \sin x_1$

Choose $P_{11} = 2P_{21}$, $\gamma = 10 P_{22}$

$P = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$, $\gamma = 10$

$P > 0$ because $2 > 0$ and $\det P = 1 > 0$

$\dot{V} = -x_2^2 - 10 x_1 \sin x_1$

Using $x_1 \sin x_1 > 0$, $-\pi < x_1 < \pi$, $x_1 \neq 0$

$\Rightarrow \dot{V} < 0$ for $x \neq 0$, $|x_1| < \pi$

$(0,0)$ is L.A.S by T. 4.1

Problem 6

$\dot{x}_1 = x_2 + \alpha x_1 (\beta^2 - x_1^2 - x_2^2)$

$\dot{x}_2 = -x_1 + \alpha x_2 (\beta^2 - x_1^2 - x_2^2) \quad \alpha, \beta > 0$

Chetaev's theorem with

$V(x) = \frac{1}{2}(x_1^2 + x_2^2)$

$\dot{V} = x_1\dot{x}_1 + x_2\dot{x}_2 = \alpha(x_1^2 + x_2^2)(\beta^2 - x_1^2 - x_2^2)$

Choose Lall $B_r$ with $0 < r < \beta$

For every $x \in U = B_r \setminus \{0\}$,

$V(x) > 0$

and since $\alpha > 0$ and $\|x\|^2 < r^2 < \beta^2$

$\dot{V} = \alpha \|x\|^2 (\beta^2 - \|x\|^2) > 0$

$V(0) = 0$ and there are points close to origin with $V(x) > 0$

By Chetaev's theorem $x = 0$ is unstable

Problem 7.

a) $V(x) = \frac{1}{2}\left(\frac{x_1^2}{1+x_1^2} + x_2\right)$

$V(x) > 0 \quad \forall x \neq 0$ and $V(0) = 0$

$\dot{V} = \frac{x_1}{(1+x_1^2)^2}\dot{x}_1 + x_2\dot{x}_2 = -\frac{6x_1^2}{(1+x_1^2)^4} - \frac{2x_2^2}{(1+x_1^2)^2} < 0 \quad \forall x \neq 0$

$V$ is strict L.F

b) $x_2 = 0$ and $|x_1| \to \infty$ : $V(x_1, 0) = \frac{1}{2}\frac{x_1^2}{1+x_1^2} \to \frac{1}{2}$

Therefore $V$ is not R.U

c) T.4.1 says the origin is A.S

T.4.2 says we can't conclude G.A.S because $V$ is not R.U

Problem 8

$|g(x)| \leq k|x|^{p+1}$

p odd, $a<0$:

$V(x) = \frac{1}{2}x^2$

$\dot{V} = x\dot{x} = ax^{p+1} + xg(x)$

$p+1$ is even,

$\dot{V} \leq -|a||x|^{p+1} + k|x|^{p+2} = -|x|^{p+1}(|a| - k|x|)$

For $|x| < \frac{|a|}{k}$ $\dot{V} < 0$ for $x \neq 0$. $x=0$ is L.A.S

p odd, $a>0$

$V(x) = \frac{1}{2}x^2$

$\dot{V} \geq |x|^{p+1}(a - k|x|) > 0$ for small $x \neq 0$

Chetaev says $x=0$ is unstable

p even, $a \neq 0$

$V(x) = \text{sgn}(a)\, x$

$V > 0$:

$\dot{V} \geq |x|^p(|a| - k|x|) > 0$ for small $x \neq 0$

Chetaev: $x=0$ is unstable

Problem 9

Equilibria:

$\dot{x} = -x_2 = 0 \Rightarrow x_2 = 0$

$\dot{x}_1 = -x_1 = 0 \Rightarrow x_1 = 0, \quad \dot{x}_3 = -x_3 = 0 \Rightarrow x_3 = 0$

$x^* = 0$ is only equilibrium

$V = \frac{1}{2} x^T P x, \quad P = \begin{bmatrix} P_{11} & P_{12} & 0 \\ P_{12} & P_{22} & 0 \\ 0 & 0 & P_{33} \end{bmatrix}$

$\dot{V} = -P_{11} x_1^2 + (2P_{11} - 2P_{12}) x_1 x_2 + (2P_{12} - P_{22}) x_2^2 - P_{33} x_3^2$
$+ (P_{33} - P_{11}) x_1 x_2 x_3 + (P_{33} - P_{12}) x_2 x_3$

$P_{11} = P_{12} = P_{33} = 1, \quad P_{22} = 3$

$P = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 3 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

$\dot{V} = -x_1^2 - x_2^2 - x_3^2 = -\|x\|^2$

$P > 0 \Rightarrow \frac{2 - \sqrt{2}}{2} \|x\|^2 \leq V(x) \leq \frac{2 + \sqrt{2}}{2} \|x\|^2$

$\dot{V} \leq -\|x\|^2 \quad \forall \; x \in \mathbb{R}^3$

Theorem 4.10 with $a = 2$:

$x = 0$ is G.A.S